



DESIGN GUIDELINES

SOCIAL ROBOT CO-DESIGN CANVASES

What things are important to consider in the robot's design?

Advantage guidelines

What advantages can the robotic solution have?
Think back to what you defined in the solution space canvas.

Prioritise advantages that reduce cognitive load during triage: rapid, consistent ESI-level prediction; plain-language explanations of contributing factors; reduced decision stress during busy shifts; and full clinician override at every step, so the AI speeds up decisions without replacing judgement.

Ethical guidelines

What ethical considerations does the robot have?
Think back to what you defined in the ethics canvas.

Preserve nurse authority over every triage decision, since an incorrect recommendation accepted uncritically could delay treatment. Keep recommendations transparent and explainable, protect patient data through encryption and role-based access, and regularly validate the model for bias across age, gender and patient groups.

ROBOT DIMENSIONS

Environment guidelines

What should the robot's context be like?

For example:

- If users are especially vulnerable, should it optimize for support?
- If the robot is part of a strict process, should it optimize for efficiency and security?

Patients are highly vulnerable and time-critical, and the ED triage process is strict, so the dashboard must optimise for both patient safety/support and operational efficiency/security. It should integrate into the existing workflow without adding documentation time or disrupting patient flow.

Form guidelines

What guides the design of the robot's outward qualities?

For example:

- Should the robot be designed to appear especially approachable, or more industrial?
- Should it be simple, or detailed?

Favour a simple, uncluttered, high-contrast interface over decorative detail: large readable text, clear icons and consistent colour-coding. The dashboard should feel approachable but clinically professional, prioritising readability and speed under stress over visual sophistication.

Interaction guidelines

What guides the design of interaction?

For example:

- Is the interaction multimodal, or is one modality optimized for efficiency?
- Should the user feel empowered and lead the interaction, or does the robot provide safety via leadership?
- Is the goal of the interaction to complete a task, or explore?

Optimise for efficiency rather than exploration: minimise clicks and typing so the nurse can move through an assessment quickly. The nurse should feel empowered and lead the interaction, with the AI providing a supporting safety net rather than directing the workflow.

Behaviour guidelines

What guides the design of the robot's behaviour?

For example:

- Should behaviour be simple, or sensitive to context?
- Does the robot have internal drivers, or does it react to external stimuli?
- Does the robot have social skills?

Behaviour should be context-sensitive, reacting to live vitals, symptoms and EHR data rather than following fixed internal rules. Communication should stay calm and consistent to avoid alarm fatigue, while still making genuinely urgent findings clearly noticeable.

